

Loops Zero to Hero Questions List

June 4, 2026

1 C Programming – for Loop Practice Questions (Only Questions with Hints)

1.1 Beginner Level

1. Print numbers from 1 to 10 using a `for` loop.
 - **Hint:** Start from 1 and increment by 1.
 2. Print numbers from 10 to 1 in reverse order.
 - **Hint:** Use decrement operator.
 3. Print all even numbers from 1 to 50.
 - **Hint:** Check `i % 2 == 0`.
 4. Print all odd numbers from 1 to 50.
 - **Hint:** Check `i % 2 != 0`.
 5. Print the first 20 natural numbers.
 - **Hint:** Loop from 1 to 20.
 6. Find the sum of numbers from 1 to N.
 - **Hint:** Maintain a running sum.
 7. Find the product of numbers from 1 to N.
 - **Hint:** Initialize product as 1.
 8. Print the multiplication table of a given number.
 - **Hint:** Multiply the number by values from 1 to 10.
 9. Print squares of numbers from 1 to N.
 - **Hint:** Use `i*i`.
 10. Print cubes of numbers from 1 to N.
 - **Hint:** Use `i*i*i`.
-

1.2 Easy Level

11. Count the number of even numbers between 1 and N.
 - **Hint:** Increment counter when number is even.
 12. Count the number of odd numbers between 1 and N.
 - **Hint:** Increment counter when number is odd.
 13. Find the sum of all even numbers from 1 to N.
 - **Hint:** Add only even numbers.
 14. Find the sum of all odd numbers from 1 to N.
 - **Hint:** Add only odd numbers.
 15. Print all multiples of 5 up to N.
 - **Hint:** Use `i % 5 == 0`.
 16. Print numbers divisible by both 3 and 5.
 - **Hint:** Use logical AND.
 17. Print the first N multiples of a given number.
 - **Hint:** Multiply loop variable by the number.
 18. Calculate the average of numbers from 1 to N.
 - **Hint:** Average = Sum / Count.
 19. Find the largest number entered among N inputs.
 - **Hint:** Track maximum value.
 20. Find the smallest number entered among N inputs.
 - **Hint:** Track minimum value.
-

1.3 Intermediate Level

21. Find the factorial of a number.
 - **Hint:** Multiply all numbers from 1 to N.
22. Check whether a number is prime.
 - **Hint:** Count divisors.
23. Print all factors of a number.
 - **Hint:** Check divisibility from 1 to N.
24. Count the factors of a number.
 - **Hint:** Increment count whenever divisible.

25. Find the sum of factors of a number.
- **Hint:** Add factors only.
26. Print all prime numbers from 1 to N.
- **Hint:** Use nested **for** loops.
27. Count prime numbers between 1 and N.
- **Hint:** Check each number for primality.
28. Find the sum of prime numbers up to N.
- **Hint:** Add primes only.
29. Print all perfect numbers up to N.
- **Hint:** Sum of proper divisors equals number.
30. Check whether a number is a perfect number.
- **Hint:** Sum all proper divisors.
-

1.4 Pattern-Based Questions

31. Print:

```
*
**
***
****
*****
```

- **Hint:** Nested **for** loops.

32. Print:

```
*****
****
***
**
*
```

- **Hint:** Decrease inner loop count.

33. Print:

```
1
12
123
1234
12345
```

- **Hint:** Print column number.

34. Print:

```
12345
1234
123
12
1
```

- **Hint:** Reverse triangle.

35. Print:

```
A
AB
ABC
ABCD
ABCDE
```

- **Hint:** Use character variables.

36. Print Floyd's Triangle.

- **Hint:** Maintain a counter variable.

37. Print a square pattern of N rows and N columns.

- **Hint:** Nested loops.

38. Print a hollow square pattern.

- **Hint:** Print * only on borders.

39. Print a right-aligned triangle.

- **Hint:** Use spaces and stars.

40. Print a pyramid pattern.

- **Hint:** Control spaces and stars separately.

1.5 Number Theory Questions

41. Find whether a number is an Armstrong number.

- **Hint:** Extract digits using loops.

42. Find whether a number is a Strong number.

- **Hint:** Sum factorials of digits.

43. Find whether a number is a Palindrome.

- **Hint:** Reverse the number.

44. Reverse a number.

- **Hint:** Use modulus and division.

45. Count digits in a number.
- **Hint:** Divide repeatedly by 10.
46. Find the sum of digits.
- **Hint:** Extract digits one by one.
47. Find the product of digits.
- **Hint:** Multiply extracted digits.
48. Find the largest digit in a number.
- **Hint:** Compare extracted digits.
49. Find the smallest digit in a number.
- **Hint:** Track minimum digit.
50. Check whether a number is a Harshad number.
- **Hint:** Number divisible by sum of digits.
-

1.6 Advanced for Loop Questions

51. Generate Fibonacci series up to N terms.
- **Hint:** Use three variables.
52. Find the Nth Fibonacci number.
- **Hint:** Update previous two terms.
53. Print Pascal's Triangle.
- **Hint:** Use combinations formula.
54. Calculate (x^n) without using `pow()`.
- **Hint:** Repeated multiplication.
55. Calculate $(1^2 + 2^2 + 3^2 + \dots + N^2)$.
- **Hint:** Sum squares.
56. Calculate $(1^3 + 2^3 + \dots + N^3)$.
- **Hint:** Sum cubes.
57. Find GCD of two numbers using loops.
- **Hint:** Check common divisors.
58. Find LCM of two numbers using loops.
- **Hint:** Find first common multiple.
59. Print all twin prime pairs up to N.

- **Hint:** Difference between primes is 2.
60. Print all prime pairs whose sum equals N.
- **Hint:** Goldbach-type approach using nested loops.
-

1.7 Expert Level

61. Print a butterfly pattern.
- **Hint:** Four sections using nested loops.
62. Print a diamond pattern.
- **Hint:** Upper and lower pyramids.
63. Find all Armstrong numbers between 1 and N.
- **Hint:** Check each number individually.
64. Find all Strong numbers between 1 and N.
- **Hint:** Factorial of digits.
65. Generate multiplication tables from 1 to N.
- **Hint:** Nested loops.
66. Find all perfect numbers between 1 and N.
- **Hint:** Sum divisors.
67. Print all Pythagorean triplets up to N.
- **Hint:** Triple nested loops.
68. Print all combinations of two dice.
- **Hint:** Nested loops from 1 to 6.
69. Print all combinations of three-digit numbers using digits 0–9.
- **Hint:** Triple nested loops.
70. Simulate a simple calendar month using loops.
- **Hint:** Use rows and columns for weeks and days.

2 C Programming – while Loop Practice Questions (Only Questions with Hints)

2.1 Beginner Level

1. Print numbers from 1 to 10 using a `while` loop.
- **Hint:** Start with 1 and increment until 10.
2. Print numbers from 10 to 1.

- **Hint:** Use a decrementing counter.
3. Print all even numbers from 1 to 50.
 - **Hint:** Check `num % 2 == 0`.
 4. Print all odd numbers from 1 to 50.
 - **Hint:** Check `num % 2 != 0`.
 5. Print the first N natural numbers.
 - **Hint:** Use a counter variable.
 6. Find the sum of numbers from 1 to N.
 - **Hint:** Add each number to a running total.
 7. Find the product of numbers from 1 to N.
 - **Hint:** Initialize product as 1.
 8. Print the multiplication table of a given number.
 - **Hint:** Counter from 1 to 10.
 9. Print squares of numbers from 1 to N.
 - **Hint:** Use `num * num`.
 10. Print cubes of numbers from 1 to N.
 - **Hint:** Use `num * num * num`.
-

2.2 Easy Level

11. Count the number of even numbers from 1 to N.
 - **Hint:** Increment count when number is even.
12. Count the number of odd numbers from 1 to N.
 - **Hint:** Increment count when number is odd.
13. Find the sum of all even numbers from 1 to N.
 - **Hint:** Add only even numbers.
14. Find the sum of all odd numbers from 1 to N.
 - **Hint:** Add only odd numbers.
15. Print all multiples of 3 up to N.
 - **Hint:** Check divisibility.
16. Print all multiples of 7 up to N.
 - **Hint:** Use modulus operator.

17. Find the average of numbers from 1 to N.
 - **Hint:** Average = Sum / Count.
 18. Find the largest among N user-entered numbers.
 - **Hint:** Keep updating maximum.
 19. Find the smallest among N user-entered numbers.
 - **Hint:** Keep updating minimum.
 20. Count how many numbers entered are positive.
 - **Hint:** Use a counter.
-

2.3 Intermediate Level

21. Find the factorial of a number.
 - **Hint:** Multiply numbers repeatedly.
 22. Check whether a number is prime.
 - **Hint:** Count divisors using a loop.
 23. Print all factors of a number.
 - **Hint:** Check divisibility from 1 to N.
 24. Count factors of a number.
 - **Hint:** Increment counter for every factor.
 25. Find the sum of factors of a number.
 - **Hint:** Add divisors only.
 26. Print all prime numbers up to N.
 - **Hint:** Use nested while loops.
 27. Count prime numbers between 1 and N.
 - **Hint:** Check each number individually.
 28. Find the sum of prime numbers up to N.
 - **Hint:** Add only primes.
 29. Check whether a number is a perfect number.
 - **Hint:** Sum proper divisors.
 30. Print all perfect numbers up to N.
 - **Hint:** Test each number.
-

2.4 Digit-Based Problems

31. Count digits in a number.
 - **Hint:** Divide repeatedly by 10.
 32. Find the sum of digits.
 - **Hint:** Use % 10 and / 10.
 33. Find the product of digits.
 - **Hint:** Multiply extracted digits.
 34. Reverse a number.
 - **Hint:** Build reversed number digit by digit.
 35. Check whether a number is a palindrome.
 - **Hint:** Compare original and reversed numbers.
 36. Find the largest digit in a number.
 - **Hint:** Compare extracted digits.
 37. Find the smallest digit in a number.
 - **Hint:** Track minimum digit.
 38. Count even digits in a number.
 - **Hint:** Check extracted digit.
 39. Count odd digits in a number.
 - **Hint:** Check extracted digit.
 40. Find the sum of even digits.
 - **Hint:** Add only even digits.
-

2.5 Number Property Problems

41. Check whether a number is an Armstrong number.
 - **Hint:** Sum powers of digits.
42. Print Armstrong numbers between 1 and N.
 - **Hint:** Test each number.
43. Check whether a number is a Strong number.
 - **Hint:** Sum factorials of digits.
44. Print Strong numbers between 1 and N.
 - **Hint:** Test each number separately.

45. Check whether a number is a Harshad number.
- **Hint:** Number divisible by sum of digits.
46. Check whether a number is a Neon number.
- **Hint:** Square the number and sum its digits.
47. Check whether a number is a Duck number.
- **Hint:** Contains zero but not as first digit.
48. Check whether a number is an Automorphic number.
- **Hint:** Square ends with the original number.
49. Check whether a number is a Disarium number.
- **Hint:** Digit powers based on positions.
50. Check whether a number is a Spy number.
- **Hint:** Sum of digits equals product of digits.
-

2.6 Pattern-Based Questions

51. Print:

```
*
**
***
****
*****
```

- **Hint:** Nested while loops.

52. Print:

```
*****
****
***
**
*
```

- **Hint:** Reduce inner loop count.

53. Print:

```
1
12
123
1234
```

- **Hint:** Use row and column counters.

54. Print Floyd's Triangle.

- **Hint:** Maintain a running number.
55. Print a square pattern.
- **Hint:** Nested loops.
56. Print a hollow square pattern.
- **Hint:** Print stars only on boundaries.
57. Print a right-aligned triangle.
- **Hint:** Use spaces and stars.
58. Print a pyramid pattern.
- **Hint:** Two nested loops.
59. Print a diamond pattern.
- **Hint:** Upper and lower halves.
60. Print Pascal's Triangle.
- **Hint:** Use combinations.
-

2.7 Input-Controlled While Loop Problems

61. Keep accepting numbers until the user enters 0.
- **Hint:** Sentinel-controlled loop.
62. Keep accepting numbers until a negative number is entered.
- **Hint:** Stop when condition fails.
63. Calculate the sum of user-entered numbers until 0 is entered.
- **Hint:** Add before checking next input.
64. Count how many numbers were entered before termination.
- **Hint:** Increment counter each iteration.
65. Find the largest number entered until user enters 0.
- **Hint:** Update maximum.
66. Find the smallest number entered until user enters 0.
- **Hint:** Update minimum.
67. Calculate average of numbers entered until 0.
- **Hint:** Maintain sum and count.
68. Count positive, negative, and zero values entered.
- **Hint:** Use three counters.

69. Simulate a simple menu-driven calculator using `while`.
- **Hint:** Repeat until Exit option selected.
70. Implement a login system with 3 attempts.
- **Hint:** Stop on success or after 3 failures.
-

2.8 Advanced While Loop Questions

71. Generate Fibonacci series up to N terms.
- **Hint:** Update previous two terms.
72. Find the Nth Fibonacci number.
- **Hint:** Repeat until N terms.
73. Calculate (x^n) without using `pow()`.
- **Hint:** Repeated multiplication.
74. Find GCD of two numbers.
- **Hint:** Test divisors using a loop.
75. Find LCM of two numbers.
- **Hint:** Search for common multiple.
76. Print all twin prime pairs up to N.
- **Hint:** Difference between primes is 2.
77. Print all prime pairs whose sum equals N.
- **Hint:** Check every pair.
78. Print all Pythagorean triplets up to N.
- **Hint:** Triple nested while loops.
79. Simulate compound interest calculation year by year.
- **Hint:** Repeat for given years.
80. Create a number guessing game.
- **Hint:** Continue until correct guess is entered.

3 C Programming – `do...while` Loop Practice Questions (Only Questions with Hints)

Note: These problems are specifically chosen where `do...while` is natural because the loop body must execute at least once.

4 Beginner Level

1. Print numbers from 1 to 10 using a `do...while` loop.
 - **Hint:** Initialize counter before the loop.
 2. Print numbers from 10 to 1.
 - **Hint:** Decrement the counter inside the loop.
 3. Print all even numbers from 1 to 50.
 - **Hint:** Check `num % 2 == 0`.
 4. Print all odd numbers from 1 to 50.
 - **Hint:** Check `num % 2 != 0`.
 5. Print the first N natural numbers.
 - **Hint:** Increment counter each iteration.
 6. Find the sum of numbers from 1 to N.
 - **Hint:** Maintain a running total.
 7. Find the product of numbers from 1 to N.
 - **Hint:** Initialize product as 1.
 8. Print the multiplication table of a given number.
 - **Hint:** Repeat 10 times.
 9. Print squares of numbers from 1 to N.
 - **Hint:** Use `num * num`.
 10. Print cubes of numbers from 1 to N.
 - **Hint:** Use `num * num * num`.
-

5 Easy Level

11. Count even numbers from 1 to N.
 - **Hint:** Use a counter variable.
12. Count odd numbers from 1 to N.
 - **Hint:** Increment count for odd values.
13. Find the sum of all even numbers from 1 to N.
 - **Hint:** Add only even numbers.
14. Find the sum of all odd numbers from 1 to N.
 - **Hint:** Add only odd numbers.

15. Print multiples of 3 up to N.
 - **Hint:** Check divisibility.
 16. Print multiples of 7 up to N.
 - **Hint:** Use modulus operator.
 17. Find the average of numbers from 1 to N.
 - **Hint:** Use sum and count.
 18. Find the largest among N entered numbers.
 - **Hint:** Update maximum whenever needed.
 19. Find the smallest among N entered numbers.
 - **Hint:** Update minimum whenever needed.
 20. Count positive numbers among N inputs.
 - **Hint:** Increment counter for positive values.
-

6 User-Controlled Input Problems

21. Accept numbers until the user enters 0.
 - **Hint:** Sentinel value = 0.
22. Find the sum of numbers entered until 0.
 - **Hint:** Add each number before checking next input.
23. Count numbers entered until 0.
 - **Hint:** Increment counter each iteration.
24. Find the largest number entered until 0.
 - **Hint:** Keep updating maximum.
25. Find the smallest number entered until 0.
 - **Hint:** Keep updating minimum.
26. Calculate average until 0 is entered.
 - **Hint:** Maintain sum and count.
27. Count positive and negative numbers until user enters 0.
 - **Hint:** Use two counters.
28. Keep accepting numbers until a negative value is entered.
 - **Hint:** Loop condition checks negativity.
29. Keep accepting passwords until the correct password is entered.

- **Hint:** Compare input with stored password.
30. Keep accepting a menu choice until Exit is selected.
- **Hint:** Repeat menu display.
-

7 Digit Manipulation Problems

31. Count digits in a number.
- **Hint:** Divide repeatedly by 10.
32. Find the sum of digits.
- **Hint:** Extract digits using % 10.
33. Find the product of digits.
- **Hint:** Multiply extracted digits.
34. Reverse a number.
- **Hint:** Build reversed number digit by digit.
35. Check whether a number is a palindrome.
- **Hint:** Compare original and reversed numbers.
36. Find the largest digit in a number.
- **Hint:** Compare extracted digits.
37. Find the smallest digit in a number.
- **Hint:** Track minimum digit.
38. Count even digits.
- **Hint:** Check extracted digit.
39. Count odd digits.
- **Hint:** Check extracted digit.
40. Find the sum of even digits.
- **Hint:** Add only even digits.
-

8 Number Property Problems

41. Check whether a number is prime.
- **Hint:** Count divisors.
42. Print all factors of a number.

- **Hint:** Check divisibility from 1 to N.
43. Count factors of a number.
- **Hint:** Increment counter whenever divisible.
44. Find the sum of factors.
- **Hint:** Add divisors only.
45. Check whether a number is perfect.
- **Hint:** Sum proper divisors.
46. Check whether a number is Armstrong.
- **Hint:** Sum powers of digits.
47. Check whether a number is Strong.
- **Hint:** Sum factorials of digits.
48. Check whether a number is Harshad.
- **Hint:** Number divisible by sum of digits.
49. Check whether a number is Automorphic.
- **Hint:** Square ends with original number.
50. Check whether a number is Neon.
- **Hint:** Sum digits of square.
-

9 Pattern Problems Using `do...while`

51. Print:

```
*
**
***
****
*****
```

- **Hint:** Nested `do...while` loops.

52. Print:

```
*****
****
***
**
*
```

- **Hint:** Reduce inner loop count.

53. Print:

1
12
123
1234

- **Hint:** Print column number.

54. Print Floyd's Triangle.

- **Hint:** Maintain running counter.

55. Print a square pattern.

- **Hint:** Nested loops.

56. Print a hollow square.

- **Hint:** Print stars on borders only.

57. Print a right-aligned triangle.

- **Hint:** Print spaces before stars.

58. Print a pyramid pattern.

- **Hint:** Separate loops for spaces and stars.

59. Print a diamond pattern.

- **Hint:** Upper and lower halves.

60. Print Pascal's Triangle.

- **Hint:** Use combination logic.

10 Menu-Driven Applications (Best Use of do...while)

61. Menu-driven calculator.

- **Hint:** Repeat until Exit option.

62. Simple ATM menu.

- **Hint:** Balance inquiry, deposit, withdraw.

63. Student result processing menu.

- **Hint:** Continue until Exit.

64. Unit conversion menu.

- **Hint:** Length, weight, temperature.

65. Area calculation menu.

- **Hint:** Circle, rectangle, triangle.

66. Number system conversion menu.

- **Hint:** Decimal to Binary, Octal, Hex.
 - 67. Banking system menu.
 - **Hint:** Deposit, withdraw, check balance.
 - 68. Employee management menu.
 - **Hint:** Add, search, display records.
 - 69. Library management menu.
 - **Hint:** Issue, return, search books.
 - 70. Restaurant ordering system.
 - **Hint:** Continue ordering until checkout.
-

11 Advanced do...while Problems

- 71. Generate Fibonacci series up to N terms.
 - **Hint:** Update previous two terms.
- 72. Find the Nth Fibonacci number.
 - **Hint:** Repeat N times.
- 73. Calculate (x^n) without `pow()`.
 - **Hint:** Repeated multiplication.
- 74. Find GCD of two numbers.
 - **Hint:** Check divisors using a loop.
- 75. Find LCM of two numbers.
 - **Hint:** Search for first common multiple.
- 76. Print all prime numbers up to N.
 - **Hint:** Check every number.
- 77. Print all Armstrong numbers up to N.
 - **Hint:** Test each number.
- 78. Print all perfect numbers up to N.
 - **Hint:** Check divisor sums.
- 79. Number guessing game.
 - **Hint:** Continue until correct guess.
- 80. Quiz application with replay option.
 - **Hint:** Ask user whether to play again after completion.

12 C Programming – Mixed Loop Practice Questions (for + while + do...while)

12.0.1 (Only Questions with Hints)

These questions are designed so that you can solve them using **any loop** (for, while, or do...while). Try solving each problem using all three loop types for maximum practice.

13 Level 1: Basic Looping

1. Print numbers from 1 to N.
 - **Hint:** Start from 1 and increment until N.
 2. Print numbers from N to 1.
 - **Hint:** Use a decreasing counter.
 3. Print all even numbers between 1 and N.
 - **Hint:** Check divisibility by 2.
 4. Print all odd numbers between 1 and N.
 - **Hint:** Use modulus operator.
 5. Print the first N natural numbers.
 - **Hint:** Counter starts from 1.
 6. Find the sum of numbers from 1 to N.
 - **Hint:** Maintain a running total.
 7. Find the product of numbers from 1 to N.
 - **Hint:** Initialize product as 1.
 8. Find the sum of even numbers from 1 to N.
 - **Hint:** Add only even numbers.
 9. Find the sum of odd numbers from 1 to N.
 - **Hint:** Add only odd numbers.
 10. Count even and odd numbers between 1 and N.
 - **Hint:** Use two counters.
-

14 Level 2: Multiples and Tables

11. Print multiplication table of a number.

- **Hint:** Repeat 10 times.
12. Print multiplication tables from 1 to N.
 - **Hint:** Use nested loops.
 13. Print multiples of 3 up to N.
 - **Hint:** Check `num % 3 == 0`.
 14. Print multiples of 5 up to N.
 - **Hint:** Use divisibility.
 15. Print numbers divisible by both 3 and 5.
 - **Hint:** Use logical AND.
 16. Count multiples of a given number within a range.
 - **Hint:** Increment count when divisible.
 17. Find the sum of multiples of 7 up to N.
 - **Hint:** Add only matching numbers.
 18. Print squares from 1 to N.
 - **Hint:** Use `i*i`.
 19. Print cubes from 1 to N.
 - **Hint:** Use `i*i*i`.
 20. Calculate the average of numbers from 1 to N.
 - **Hint:** Average = Sum / Count.
-

15 Level 3: Input-Based Problems

21. Find the largest among N entered numbers.
 - **Hint:** Track maximum value.
22. Find the smallest among N entered numbers.
 - **Hint:** Track minimum value.
23. Count positive numbers among N inputs.
 - **Hint:** Increment positive counter.
24. Count negative numbers among N inputs.
 - **Hint:** Increment negative counter.
25. Find the sum of N entered numbers.
 - **Hint:** Add each input.

26. Find the average of N entered numbers.
- **Hint:** Use sum and count.
27. Count how many numbers are even.
- **Hint:** Check parity.
28. Count how many numbers are odd.
- **Hint:** Check parity.
29. Find the largest and smallest in one pass.
- **Hint:** Update both values simultaneously.
30. Count occurrences of a specific number in N inputs.
- **Hint:** Compare each input.
-

16 Level 4: Factorial and Power

31. Find factorial of a number.
- **Hint:** Multiply sequential numbers.
32. Calculate x^n without using `pow()`.
- **Hint:** Repeated multiplication.
33. Calculate nCr using factorials.
- **Hint:** Formula uses three factorials.
34. Calculate nPr using factorials.
- **Hint:** Use factorial relation.
35. Find sum of factorial series:
- **Hint:** $1! + 2! + 3! + \dots + N!$
36. Find product of factorials.
- **Hint:** Multiply factorial values.
37. Compute exponential series.
- **Hint:** Use factorial denominator.
38. Compute power series sum.
- **Hint:** Repeated powers.
39. Calculate compound growth over N years.
- **Hint:** Repeat percentage increase.
40. Simulate repeated doubling of a value.

- **Hint:** Multiply by 2 each iteration.
-

17 Level 5: Number Properties

41. Check whether a number is prime.
 - **Hint:** Count divisors.
 42. Print all factors of a number.
 - **Hint:** Check divisibility.
 43. Count factors of a number.
 - **Hint:** Increment factor count.
 44. Find sum of factors.
 - **Hint:** Add divisors only.
 45. Check whether a number is perfect.
 - **Hint:** Sum proper divisors.
 46. Check whether a number is Armstrong.
 - **Hint:** Sum powers of digits.
 47. Check whether a number is Strong.
 - **Hint:** Sum factorials of digits.
 48. Check whether a number is Harshad.
 - **Hint:** Number divisible by sum of digits.
 49. Check whether a number is Automorphic.
 - **Hint:** Square ends with original number.
 50. Check whether a number is Neon.
 - **Hint:** Sum digits of square.
-

18 Level 6: Digit Manipulation

51. Count digits in a number.
 - **Hint:** Divide by 10 repeatedly.
52. Find sum of digits.
 - **Hint:** Use % 10.
53. Find product of digits.

- **Hint:** Multiply extracted digits.
 - 54. Reverse a number.
 - **Hint:** Build reverse digit by digit.
 - 55. Check whether a number is palindrome.
 - **Hint:** Compare original and reversed.
 - 56. Find largest digit.
 - **Hint:** Compare extracted digits.
 - 57. Find smallest digit.
 - **Hint:** Track minimum digit.
 - 58. Count even digits.
 - **Hint:** Check each extracted digit.
 - 59. Count odd digits.
 - **Hint:** Check each extracted digit.
 - 60. Find frequency of a specific digit.
 - **Hint:** Compare extracted digits.
-

19 Level 7: Series Problems

- 61. Generate Fibonacci series up to N terms.
 - **Hint:** Use previous two terms.
- 62. Find Nth Fibonacci number.
 - **Hint:** Update values repeatedly.
- 63. Print arithmetic progression.
 - **Hint:** Add common difference.
- 64. Print geometric progression.
 - **Hint:** Multiply by common ratio.
- 65. Find sum of arithmetic series.
 - **Hint:** Add terms one by one.
- 66. Find sum of geometric series.
 - **Hint:** Add generated terms.
- 67. Print triangular numbers.
 - **Hint:** Running sum of natural numbers.

68. Print square number series.
- **Hint:** Generate n^2 .
69. Print cube number series.
- **Hint:** Generate n^3 .
70. Print alternating sign series.
- **Hint:** Change sign every iteration.
-

20 Level 8: Pattern Problems

71. Print a square star pattern.
- **Hint:** Nested loops.
72. Print a rectangle pattern.
- **Hint:** Rows and columns.
73. Print a right triangle.
- **Hint:** Increase stars per row.
74. Print an inverted triangle.
- **Hint:** Decrease stars per row.
75. Print a pyramid.
- **Hint:** Spaces + stars.
76. Print an inverted pyramid.
- **Hint:** Reverse pyramid logic.
77. Print a diamond.
- **Hint:** Upper and lower pyramids.
78. Print Floyd's Triangle.
- **Hint:** Running number counter.
79. Print Pascal's Triangle.
- **Hint:** Use combinations.
80. Print a butterfly pattern.
- **Hint:** Four symmetric sections.
-

21 Level 9: Menu-Driven Programs

81. Calculator using menu.
 - **Hint:** Repeat until Exit.
 82. ATM simulation.
 - **Hint:** Deposit, withdraw, balance.
 83. Student result processing.
 - **Hint:** Menu loop.
 84. Area calculator.
 - **Hint:** Circle, rectangle, triangle.
 85. Unit converter.
 - **Hint:** Length, weight, temperature.
 86. Banking system.
 - **Hint:** Multiple menu options.
 87. Employee management system.
 - **Hint:** Menu repetition.
 88. Library management system.
 - **Hint:** Add, search, display.
 89. Quiz application.
 - **Hint:** Repeat if user chooses.
 90. Restaurant ordering system.
 - **Hint:** Continue until checkout.
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22 Level 10: Advanced Logic Problems

91. Print all prime numbers up to N.
 - **Hint:** Check each number for primality.
92. Print all Armstrong numbers up to N.
 - **Hint:** Test every number.
93. Print all perfect numbers up to N.
 - **Hint:** Sum divisors for each number.
94. Print all Strong numbers up to N.
 - **Hint:** Factorial of digits.

95. Print all twin prime pairs up to N.
 - **Hint:** Difference between primes is 2.
96. Find GCD of two numbers.
 - **Hint:** Check common divisors.
97. Find LCM of two numbers.
 - **Hint:** Find first common multiple.
98. Print all Pythagorean triplets up to N.
 - **Hint:** Triple nested loops.
99. Number guessing game.
 - **Hint:** Repeat until correct guess.
100. Simulate a simple calendar month.
 - **Hint:** Use rows (weeks) and columns (days).

23 C Programming – Loops in Real-Life Scenarios

23.0.1 (Using for, while, do...while, Nested Loops, or Combinations of Them)

23.0.2 Only Questions with Hints

24 Banking and Finance

1. Simulate an ATM that allows multiple transactions until the user chooses Exit.
 - **Hint:** Menu-driven program using do...while.
2. Calculate monthly savings over N months.
 - **Hint:** Add savings each month using a loop.
3. Compute compound interest year by year.
 - **Hint:** Update balance every iteration.
4. Track daily expenses for 30 days and find the total.
 - **Hint:** Use a loop to collect daily values.
5. Find the highest transaction amount among N transactions.
 - **Hint:** Maintain a maximum value.
6. Calculate EMI payments for each month of a loan.
 - **Hint:** Repeat for loan duration.
7. Count how many transactions exceed 10,000.

- **Hint:** Increment counter when condition is true.
 - 8. Simulate recurring deposits for N months.
 - **Hint:** Add monthly deposit and interest.
 - 9. Generate a yearly account statement month-wise.
 - **Hint:** Loop through 12 months.
 - 10. Simulate a simple stock investment growth over N years.
 - **Hint:** Update investment annually.
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25 Education and Examination Systems

- 11. Enter marks for N students and calculate class average.
 - **Hint:** Sum marks and divide by N.
 - 12. Find the topper among N students.
 - **Hint:** Track highest score.
 - 13. Count students who passed.
 - **Hint:** Check pass criteria.
 - 14. Generate report cards for N students.
 - **Hint:** Loop through student records.
 - 15. Calculate attendance percentage for a semester.
 - **Hint:** Count attended classes.
 - 16. Display multiplication tables for all numbers from 1 to 20.
 - **Hint:** Nested loops.
 - 17. Simulate an online quiz until the student chooses to stop.
 - **Hint:** Use `do...while`.
 - 18. Calculate GPA from N subjects.
 - **Hint:** Sum grade points.
 - 19. Count how many students scored above average.
 - **Hint:** Compare each mark with average.
 - 20. Generate seat numbers for examination halls.
 - **Hint:** Nested loops for rows and columns.
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26 Retail and Shopping

21. Calculate the total bill for N purchased items.
 - **Hint:** Add item prices.
 22. Apply discounts to all items in a shopping cart.
 - **Hint:** Loop through items.
 23. Count products that are out of stock.
 - **Hint:** Check inventory quantity.
 24. Generate an itemized invoice.
 - **Hint:** Print item details in a loop.
 25. Simulate a supermarket billing system.
 - **Hint:** Continue adding items until checkout.
 26. Find the most expensive item purchased.
 - **Hint:** Track maximum price.
 27. Calculate GST for all items.
 - **Hint:** Apply tax formula repeatedly.
 28. Count how many products belong to a certain category.
 - **Hint:** Compare category IDs.
 29. Display inventory stock levels.
 - **Hint:** Loop through product list.
 30. Simulate a self-checkout machine.
 - **Hint:** Repeat until payment.
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27 Transportation and Travel

31. Calculate total distance traveled over N trips.
 - **Hint:** Sum trip distances.
32. Find the longest trip among N journeys.
 - **Hint:** Track maximum distance.
33. Generate bus seat numbers.
 - **Hint:** Nested loops.
34. Simulate a ticket booking system.
 - **Hint:** Repeat until seats are filled.

35. Calculate fuel consumption over N days.
 - **Hint:** Sum daily fuel usage.
 36. Track vehicle speed readings every minute.
 - **Hint:** Loop through readings.
 37. Find average travel time for N trips.
 - **Hint:** Sum times and divide.
 38. Count delayed trains from a schedule.
 - **Hint:** Check delay status.
 39. Generate a travel itinerary day-wise.
 - **Hint:** Loop through days.
 40. Simulate traffic signal cycles continuously.
 - **Hint:** Infinite loop with conditions.
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28 Employee and HR Management

41. Calculate salaries for N employees.
 - **Hint:** Loop through employee records.
42. Find the highest-paid employee.
 - **Hint:** Track maximum salary.
43. Calculate total payroll expenditure.
 - **Hint:** Sum all salaries.
44. Count employees working overtime.
 - **Hint:** Compare hours worked.
45. Generate employee IDs.
 - **Hint:** Sequential numbering.
46. Track attendance for a month.
 - **Hint:** Loop through days.
47. Calculate performance bonuses.
 - **Hint:** Apply bonus formula.
48. Count employees in each department.
 - **Hint:** Use counters.
49. Generate a weekly shift schedule.

- **Hint:** Nested loops.
50. Simulate an employee login system with limited attempts.
- **Hint:** while or do...while.
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29 Healthcare and Hospitals

51. Record temperatures of N patients.
- **Hint:** Store and process readings.
52. Find the patient with the highest temperature.
- **Hint:** Track maximum value.
53. Calculate average blood pressure readings.
- **Hint:** Sum readings.
54. Count patients with fever.
- **Hint:** Compare against threshold.
55. Generate appointment slots.
- **Hint:** Loop through time intervals.
56. Simulate patient queue processing.
- **Hint:** Process until queue is empty.
57. Calculate medicine dosage schedule for N days.
- **Hint:** Repeat daily schedule.
58. Count available hospital beds.
- **Hint:** Check occupancy status.
59. Generate patient IDs.
- **Hint:** Sequential numbering.
60. Simulate hospital billing.
- **Hint:** Add charges repeatedly.
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30 Manufacturing and Engineering

61. Track machine production every hour.
- **Hint:** Sum hourly production.
62. Find the hour with maximum production.

- **Hint:** Track maximum output.
 - 63. Calculate total units produced in a week.
 - **Hint:** Loop through days.
 - 64. Count defective products in a batch.
 - **Hint:** Check quality status.
 - 65. Generate serial numbers for products.
 - **Hint:** Sequential numbering.
 - 66. Simulate assembly line operations.
 - **Hint:** Process items one by one.
 - 67. Calculate machine maintenance intervals.
 - **Hint:** Repeat after fixed cycles.
 - 68. Monitor sensor readings every minute.
 - **Hint:** Continuous loop.
 - 69. Generate inspection reports.
 - **Hint:** Loop through products.
 - 70. Calculate energy consumption for N machines.
 - **Hint:** Sum individual consumptions.
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31 Mobile Apps and Social Media

- 71. Count likes on N posts.
 - **Hint:** Sum likes.
- 72. Find the most-liked post.
 - **Hint:** Track maximum likes.
- 73. Generate notifications for N users.
 - **Hint:** Loop through users.
- 74. Simulate login attempts.
 - **Hint:** Allow limited retries.
- 75. Count active users in a day.
 - **Hint:** Check activity status.
- 76. Generate a chat history display.
 - **Hint:** Loop through messages.

77. Simulate friend requests processing.
- **Hint:** Continue until no requests remain.
78. Count followers gained each day.
- **Hint:** Sum daily increases.
79. Generate weekly usage statistics.
- **Hint:** Loop through days.
80. Simulate an app menu system.
- **Hint:** Use `do...while`.
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32 E-Commerce and Web Applications

81. Process N customer orders.
- **Hint:** Loop through order list.
82. Count completed and pending orders.
- **Hint:** Use separate counters.
83. Generate order IDs.
- **Hint:** Sequential numbering.
84. Simulate product search until item found.
- **Hint:** Use `while`.
85. Display all available products.
- **Hint:** Iterate product list.
86. Calculate total revenue from orders.
- **Hint:** Sum order values.
87. Find the best-selling product.
- **Hint:** Track highest sales count.
88. Generate shipping labels.
- **Hint:** Loop through shipments.
89. Simulate shopping cart operations.
- **Hint:** Menu-driven loop.
90. Process customer feedback forms.
- **Hint:** Repeat until all forms are handled.
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33 Advanced Real-Life Simulations

91. Simulate an elevator moving between floors.
 - **Hint:** Increment or decrement floor number.
92. Simulate a digital clock for 24 hours.
 - **Hint:** Nested loops for hours, minutes, seconds.
93. Simulate a traffic intersection.
 - **Hint:** Repeat signal cycle.
94. Simulate a voting system.
 - **Hint:** Accept votes until polling ends.
95. Simulate a parking lot management system.
 - **Hint:** Track available spaces.
96. Generate a cinema seating arrangement.
 - **Hint:** Nested loops for rows and seats.
97. Simulate a call center queue.
 - **Hint:** Process callers one by one.
98. Simulate a warehouse inventory update system.
 - **Hint:** Loop through products.
99. Simulate a library book issue/return system.
 - **Hint:** Menu-driven loop.
100. Simulate a complete university management system (students, courses, attendance, marks).
 - **Hint:** Use multiple nested loops and menus.